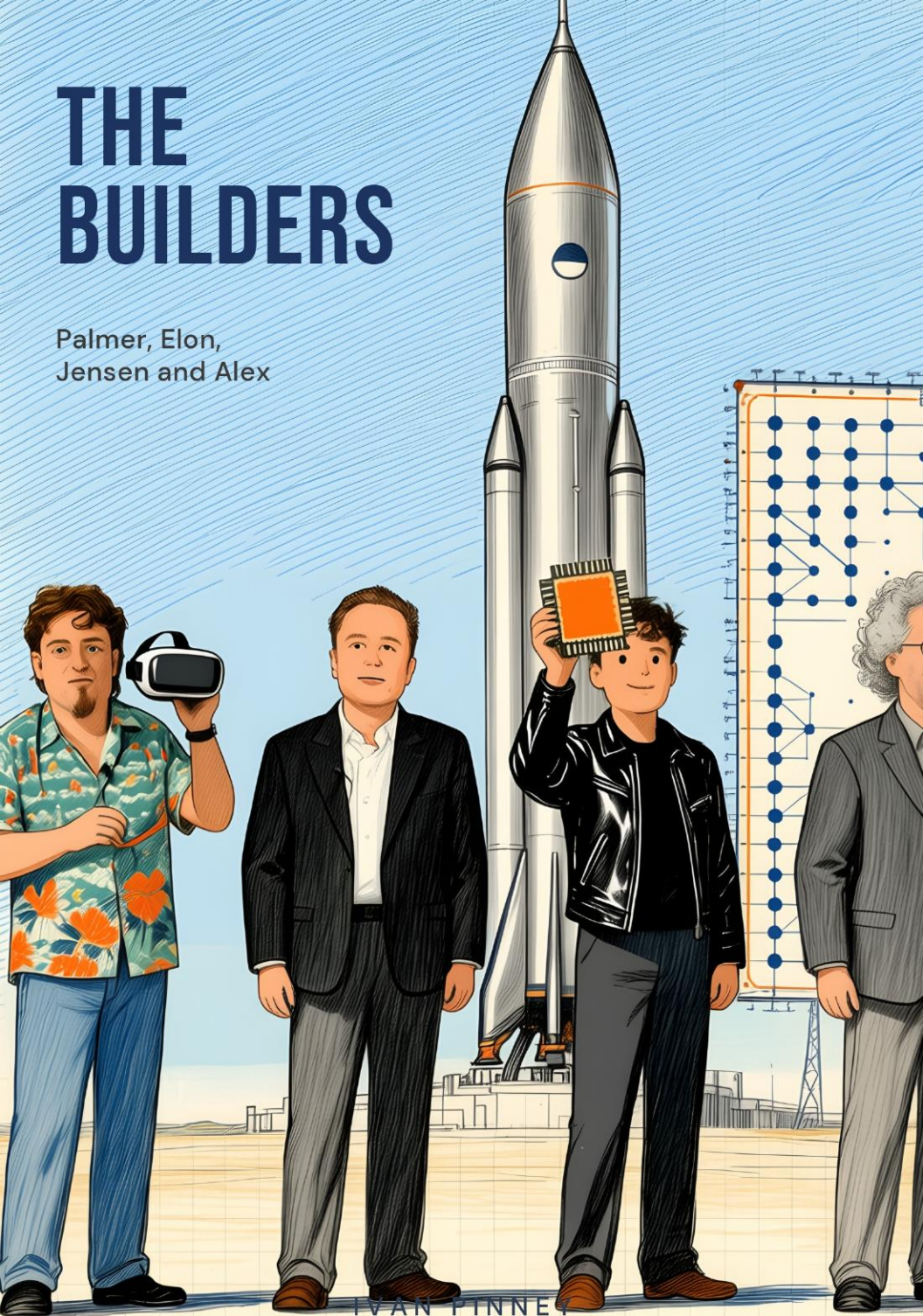


THE BUILDERS

Palmer, Elon,
Jensen and Alex



IVAN PINNEL

BUILDERS

The Builders

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Ivan Pinney

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What Is a Builder?



Look around you. Look at your chair. Look at the light over your head.

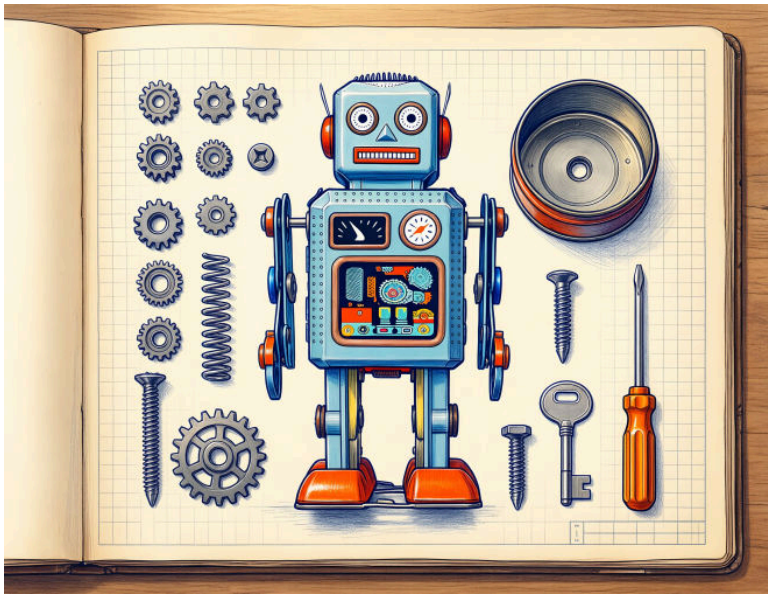
Somebody built each one.

Once, every one of those things was just an idea in a person's head. Then that person made it real.

That person was a builder.

A builder sees a problem. Then a builder asks, "How does this work?" and "How could it work better?"

Builders take things apart to see what is inside. They learn how each piece works, all the way down to the smallest part. Grown-ups call this thinking from first principles. It means you start with what is true, not with what everyone says.



Then a builder comes up with an idea to make the world better. And they get to work.

Here is a secret. You do not have to wait until you are grown up. You do not need anyone to say yes. You can just do things.

You can be anything you want to be. You can build anything you can dream up.

In this book, you will meet four real builders.



Palmer built a video game headset in his garage. Now he builds robot jets.

Elon builds the biggest rockets ever made.

Jensen makes super fast computer chips.

Alex makes computer programs that help people find what matters in giant piles of information.

Each of them was once a kid, just like you.

Builder's Toolbox: asking "How does this work?"

Let's go meet them.

Palmer: The Garage Inventor



Palmer Luckey grew up in Long Beach, California. It is a sunny city by the ocean.

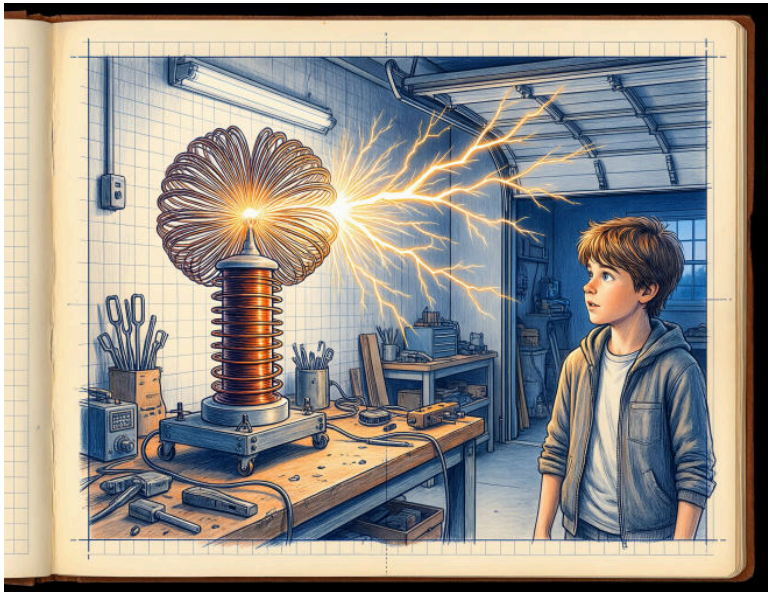
Palmer did not go to a regular school. His mom taught him at home, along with his three little sisters.

One day his parents gave him something amazing. Half of the family garage! It became Palmer's workshop.

Palmer always wanted to know how things work. So he took them apart. He wanted to see why each piece was made the way it was.

He taught himself electronics. Electronics are the wires and tiny parts that make gadgets go.

In his garage, Palmer built lasers. He built Tesla coils. A Tesla coil is a machine that makes zaps of electricity jump through the air, like tiny lightning.



Building things costs money. So Palmer found broken iPhones, fixed them and sold them. He used the money to buy more parts.

He also had a job at a sailing center. He scrubbed boat decks and fixed boat engines.

The Big Idea

When Palmer was 17, he built his first virtual reality headset. Virtual reality, or VR, is goggles that make you feel like you are inside a video game.

Palmer wanted to make it even better. So he moved into a little camper trailer in his parents' driveway. It was only 19 feet long. A computer with six screens sat at one end. His mattress sat at the other. He ate frozen burritos and kept building.



In 2012, Palmer started a company called Oculus to make his headsets. He asked people online to help. They gave \$2.4 million! In 2014, Facebook bought Oculus.

Then Palmer started something new. In 2017, he started a company called Anduril. Anduril makes robot planes and drones that help protect soldiers.

What Anduril Builds Now

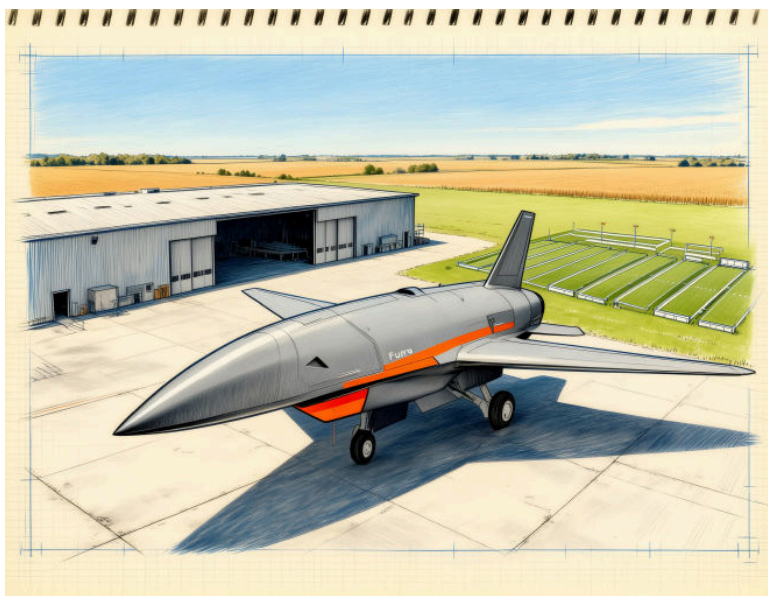
Meet Fury. Fury is a robot jet with no pilot inside. It flies next to fighter planes, like a helper. Fury flew for the first time in November 2025. In July 2026, it flew with F-35 fighter jets.



Meet Roadrunner. Roadrunner is a drone, a flying robot that works with no pilot on board.

Remember Palmer's first headset? Anduril makes a headset now too. It is called EagleEye, and it is made for soldiers.

All of this needs a giant place to build it. So Anduril built Arsenal-1, a huge factory near Columbus, Ohio. It started making things in March 2026. When it is finished, it will be as big as about 87 football fields. It is built to make up to 150 Fury jets a year.



Wow!

His Oculus helpers gave \$2.4 million.

Arsenal-1 is planned to be as big as about 87 football fields.

It can make up to 150 robot jets a year.

Builder's Toolbox: curiosity. Palmer took things apart to learn how they work.

Palmer started with half a garage and a lot of questions. A big idea can start in a small space.

Elon: The Rocket Builder



Before Elon built real rockets, he made a space game.

He was 12 years old. He had taught himself to write computer code. Code is the list of instructions that tells a computer what to do.

His game was called Blastar. It was a game about flying in space. Elon sold Blastar to a magazine for about \$500.

Twelve years old, and he had already built something people paid for.

A Boy Who Loved Books

Elon Musk was born in 1971 in Pretoria, South Africa. South Africa is a country at the very bottom tip of Africa. He grew up with a younger brother, Kimbal, and a younger sister, Tosca.

Elon loved to read. He read big, thick books full of adventure. He read *The Lord of the Rings*, a story about a long and dangerous quest. He read the *Foundation* books, stories about people far in the future, living out among the stars.

When he ran out of books, he read the encyclopedia. An encyclopedia is a giant set of books with facts about almost everything.



When Elon was about 10, he got his first computer. It came with a book about how to write code. Elon read it and taught himself. Two years later, he made Blaster.

Off on His Own

When Elon was 17, he moved to Canada. He went by himself. Canada is far from South Africa, across a whole ocean.

Think about that. A new country, on his own, at 17.

Later he went to college at the University of Pennsylvania. He studied physics. Physics is the science of how things move and why. It explains why a ball falls down and how a rocket goes up.

His First Companies

In 1995, Elon and his brother Kimbal started a company called Zip2. Zip2 helped newspapers put maps and directions on the internet. Back then, that was brand new.

In 1999, a big computer company bought Zip2.

That same year, Elon started another company. It grew into PayPal. PayPal lets people pay for things on the internet, safely and fast. In 2002, the company eBay bought PayPal.

Now Elon could do anything he wanted. He picked the hardest thing he could think of. Rockets.

The Rocket Company

In 2002, Elon started a company called SpaceX. SpaceX builds rockets. Elon wants SpaceX to help people one day live on Mars.

A rocket is a giant tube full of fuel. When the fuel burns, fire blasts out the bottom. The rocket pushes down hard, so it shoots up, fast enough to reach space.

SpaceX's first rocket was called Falcon 1.

The first time Falcon 1 launched, it failed.

The second time, it failed.

The third time, it failed again.

SpaceX had only enough money left to try one more time.

On September 28, 2008, Falcon 1 flew a fourth time. It made it all the way to orbit! Orbit means going around the Earth, high up in space.



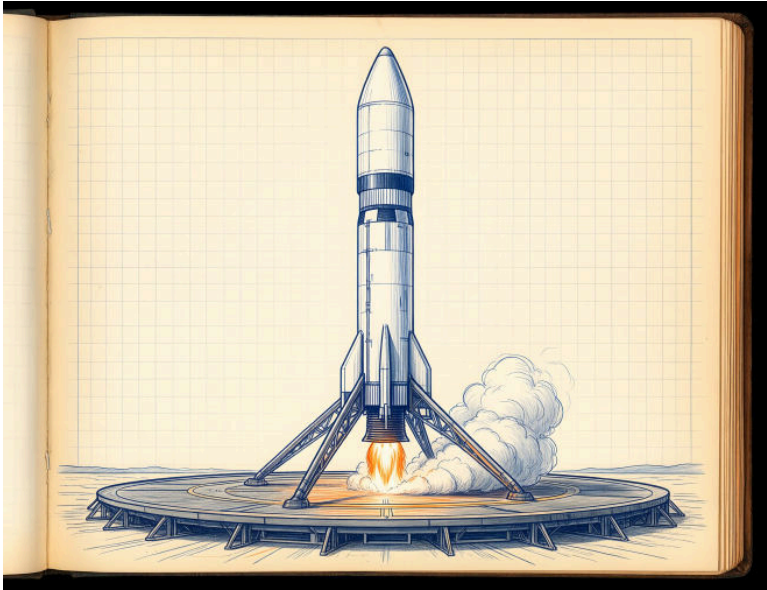
Elon did not quit after one failure, or two, or three. He kept building.

Rockets That Come Back

Next, SpaceX built a bigger rocket called Falcon 9.

The bottom part of a rocket is called the booster. The booster gives the big push to get off the ground. For a long time, boosters fell into the ocean after one flight and were never used again.

SpaceX wanted boosters that come back. In December 2015, a Falcon 9 booster flew back down and landed standing straight up. Now SpaceX boosters land again and again. Some have flown more than 20 times!



SpaceX also built a space capsule called Dragon. A capsule is the part of a spaceship where people ride. In 2020, Dragon carried two astronauts to the International Space Station. That is a giant science lab that circles the Earth.

Meet Starship

Now SpaceX is building Starship. Starship is the biggest rocket ever built.

Starship is about 408 feet tall. The Statue of Liberty, standing on its base, is 305 feet tall. Starship is taller!



Starship has two parts. The bottom part is the booster. The top part is called the ship. The ship keeps going, all the way to space.

The booster has 33 engines. Thirty-three! Picture them all roaring at once.

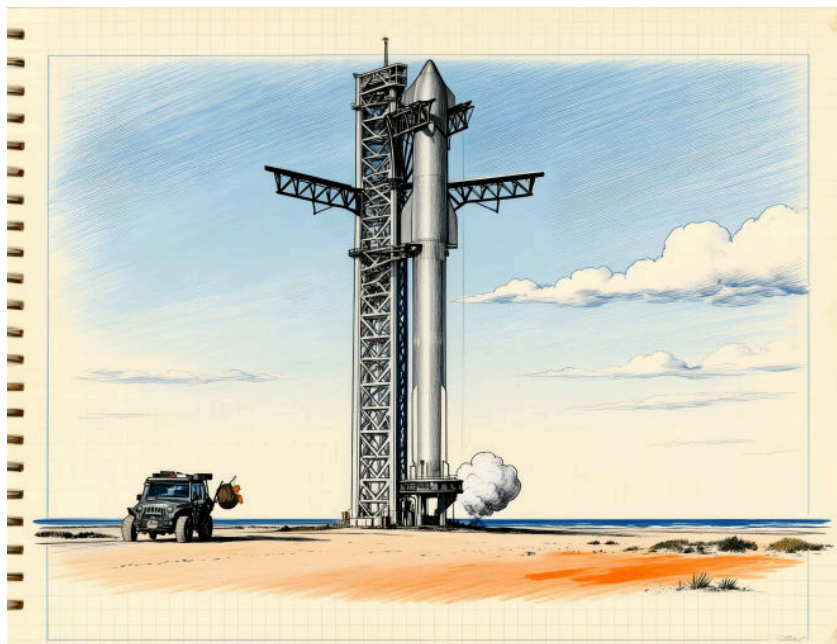
NASA picked Starship to carry astronauts to the Moon. Elon hopes it will one day carry people to Mars.

Catching a Rocket

After the Starship booster does its job, it flies back down.

On October 13, 2024, something amazing happened. The booster came back to the launch tower. The tower reached out two giant metal arms and caught it!

People call those arms the chopsticks.



Have you ever tried to pick something up with chopsticks? Now picture catching a rocket booster that way.

Around the Earth

On September 28, 2026, Starship reached orbit for the first time. That was exactly 18 years after Falcon 1 first reached orbit, on the very same day.

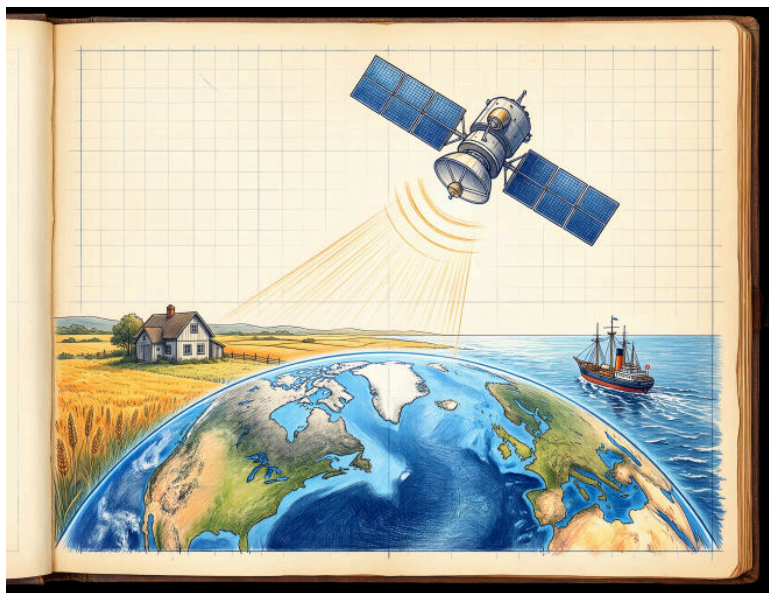
Up in orbit, Starship let out 26 new Starlink satellites. Then it came back down and splashed into the Pacific Ocean.

Starlink

What is a satellite? A satellite is a machine that circles the Earth up in space.

SpaceX has put more than 11,000 Starlink satellites up there. They beam the internet down to the ground.

Most homes get the internet through long cables. But some places have no cables. Think of a farm far from any town. Think of a ship in the middle of the ocean. Starlink beams the internet right to them, straight from space.



Tesla

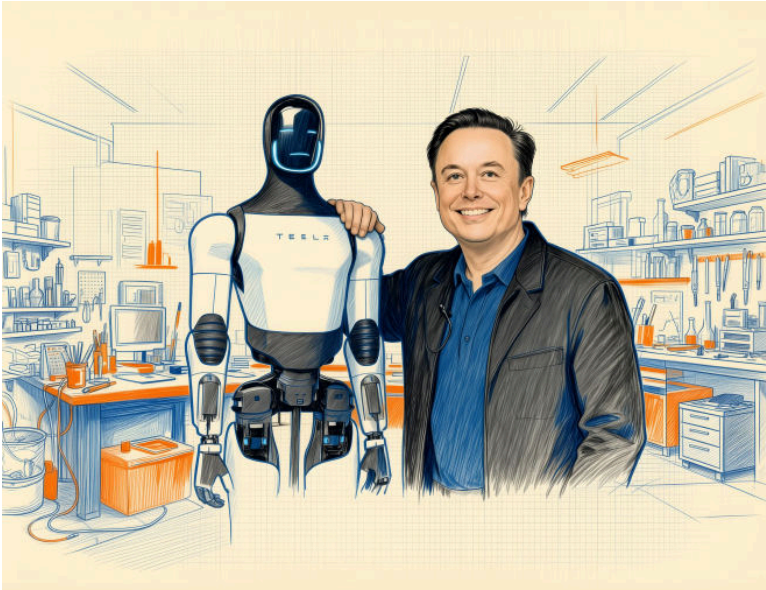
Elon builds more than rockets. In 2004, he joined a car company called Tesla.

Tesla makes electric cars. An electric car runs on electricity stored in a big battery, not on gas. You plug it in to charge it, a little like a tablet.

Tesla's first car was a sports car called the Roadster. It came out in 2008. Now Tesla makes many kinds of cars. One is the Cybertruck, a pickup truck with flat, shiny steel sides.

Tesla also makes big batteries for houses. They store electricity so the lights can stay on.

And Tesla is building a robot. Its name is Optimus. Optimus is shaped like a person, with arms, legs and a head. It is about 5 feet 8 inches tall. That is about as tall as a grown-up.



Tesla is building factories to make Optimus robots.

Neuralink

In 2016, Elon helped start a company called Neuralink. Neuralink makes a tiny device about the size of a coin.

Some people cannot move their bodies. Neuralink's device helps them use a computer just by thinking.

In 2024, the first person to get one played chess on a computer. He moved the pieces just by thinking!

The Boring Company

Elon has a company that digs tunnels. It is called The Boring Company. Boring is a word for drilling a hole.

The Boring Company digs tunnels under cities, so cars can drive underground and skip the traffic. In Las Vegas, cars carry people through tunnels under a giant building.



xAI

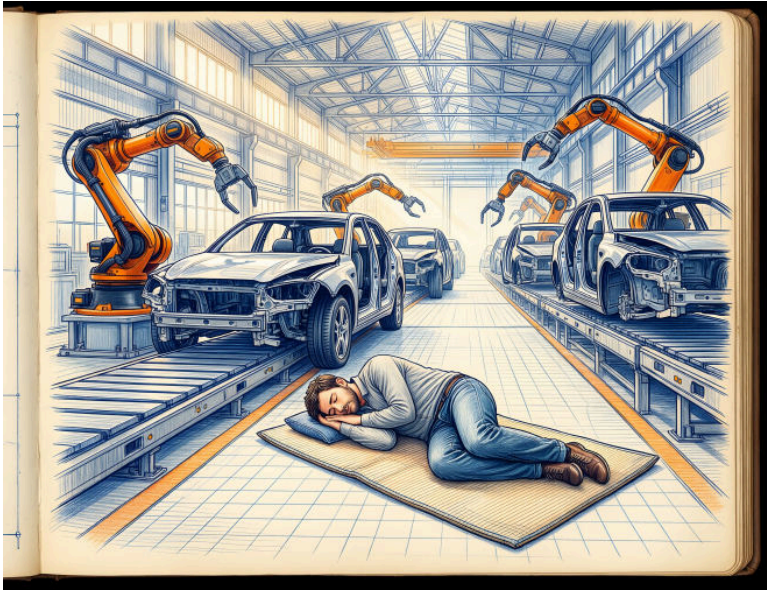
In 2023, Elon started a company called xAI. xAI builds artificial intelligence, or AI. AI is computer programs that can learn and answer questions. xAI's AI is called Grok.

AI needs a lot of computer power. So xAI built a giant computer called Colossus, in Memphis, Tennessee. It is packed with super fast computer chips made by Nvidia. You will meet Jensen, who helped start Nvidia, in the next chapter.

How Hard He Works

Building all of this takes a lot of work.

In 2018, the Tesla factory was having trouble making cars. Elon wanted to fix it. So he slept at the factory.



Why not go home at night? He said it was "Because I don't have time to go home and shower."

Wow!

Starship is about 408 feet tall, taller than the Statue of Liberty.

Its booster has 33 engines.

More than 11,000 Starlink satellites circle the Earth.

Some SpaceX boosters have flown more than 20 times.

Builder's Toolbox: never quitting. Falcon 1 failed three times. The fourth time, it flew.

At 12, Elon made a game about space. Now he builds rockets that really go there. When something fails, try again.

Jensen: From Dishwasher to Chip Maker

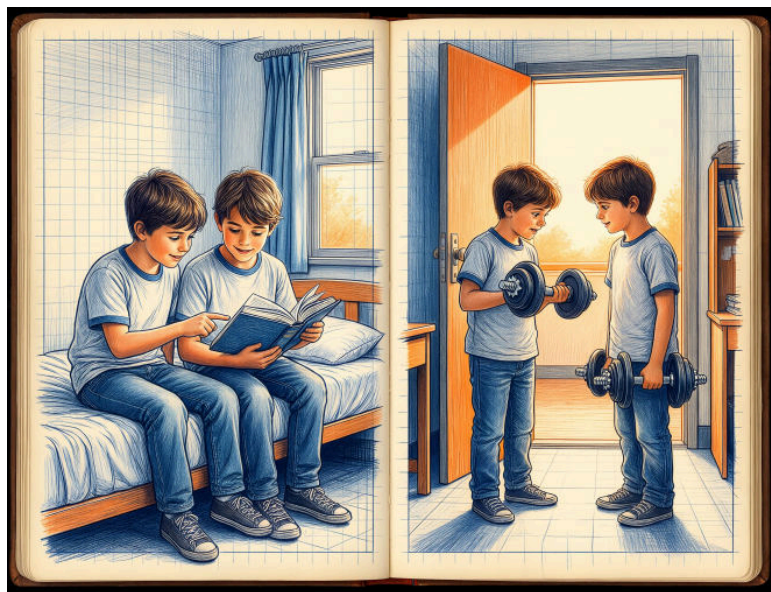


Jensen Huang was 9 years old when he came to America. He spoke very little English.

Jensen was born in 1963 in Taipei, Taiwan. Taiwan is an island in Asia. When he was 5, his family moved to Thailand. At 9, he was sent across the world to America.

He went to a school in Kentucky. Every student there had a chore. Jensen's chore was cleaning the bathrooms.

Jensen taught his roommate how to read. His roommate taught Jensen how to lift weights. Each one helped the other get stronger.



Jensen joined the swim team. He did so well in school that he skipped two grades. He finished high school in Oregon when he was only 16.

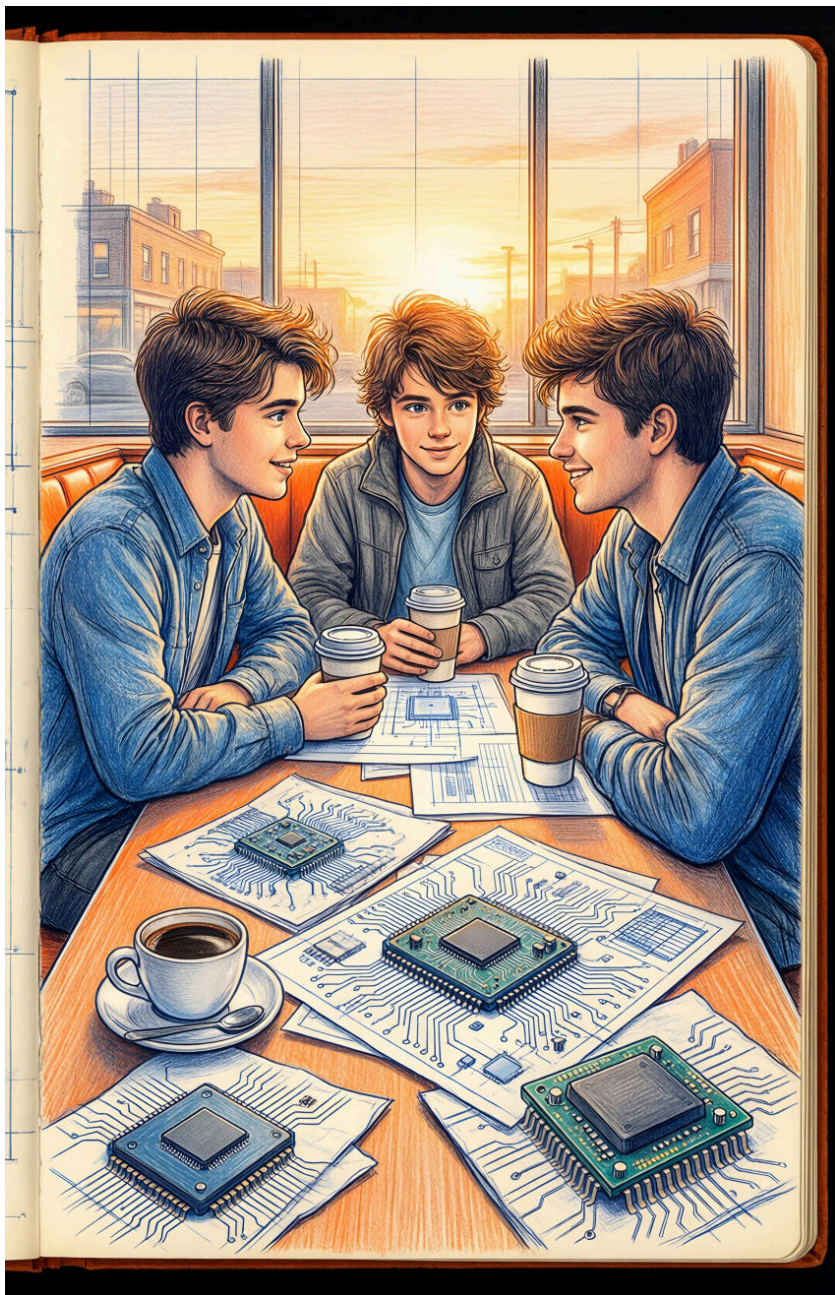
When he was 15, Jensen got a job at Denny's, a restaurant. First he washed dishes. Then he was a busboy, clearing the tables. Then he became a waiter.



Later he went to college at Oregon State. In 1984 he finished with a degree in engineering. Engineering means using math and science to design and build things.

The Big Idea

In 1993, Jensen and two friends had an idea for a new company. Where did they plan it? In a booth at a Denny's!



They called their company Nvidia.

Nvidia makes GPUs. A GPU is a special computer chip. A computer chip is a tiny part that does a computer's thinking.

A GPU is very fast at doing many little jobs at once. Imagine coloring a giant picture all by yourself. It would take forever. Now imagine your whole class coloring it, each kid doing one small square. That is how a GPU works.



At first, Nvidia's chips made video game pictures look amazing.

Now they do something even bigger. They power artificial intelligence, or AI.

What Nvidia Builds Now

Nvidia's newest chips are called Blackwell and Vera Rubin. In 2026, Vera Rubin chips are in full production. That means they are being made in huge numbers.

These chips go into data centers. A data center is a giant building full of computers, row after row after row.



How Jensen Works

Jensen never forgot where he started. He said, "To me, no task is beneath me because, remember, I used to be a dishwasher."

Jensen does not wear a watch. Why? Because "now is the most important time."

Wow!

In three months in 2026, Nvidia sold about \$96 billion of its products.

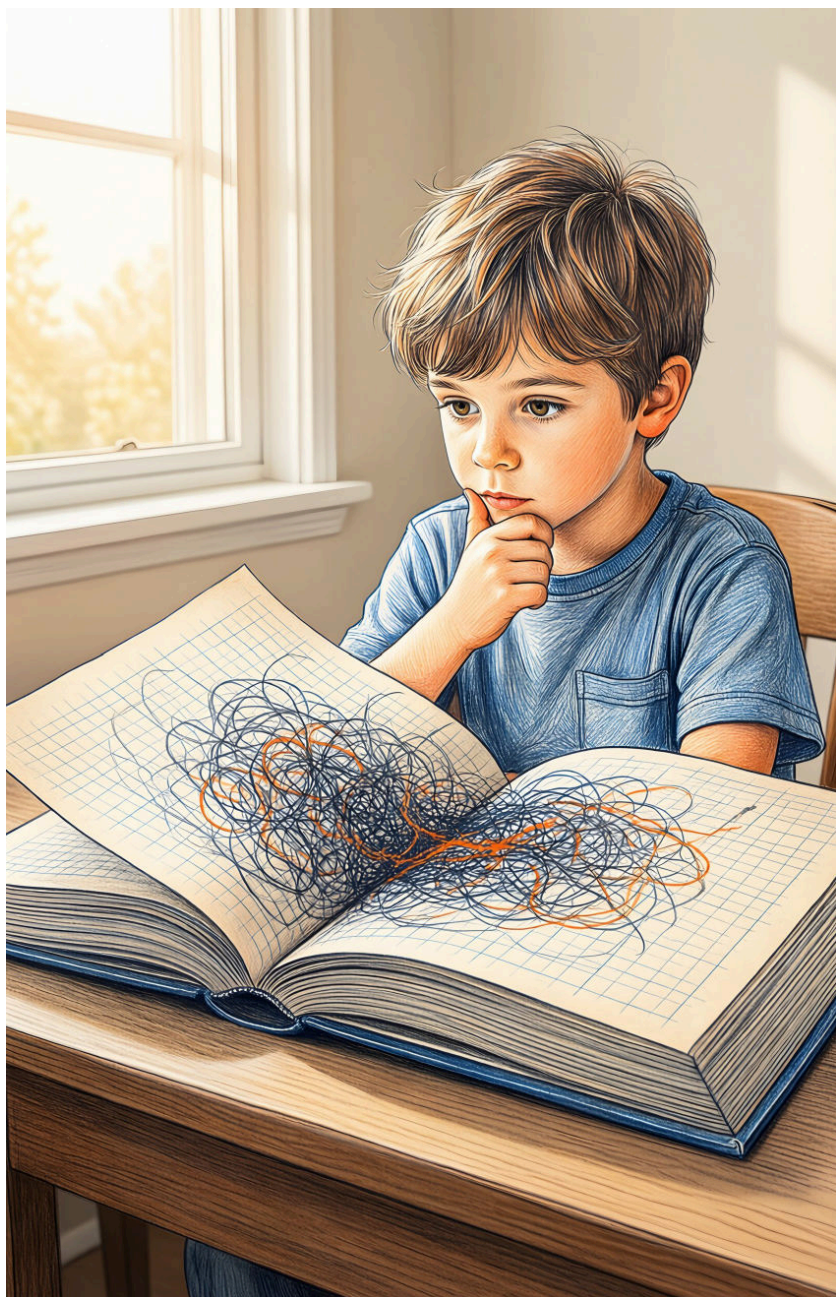
That is about \$1 billion every single day.

Nvidia is one of the most valuable companies in the world.

Builder's Toolbox: no job too small. Jensen cleaned bathrooms and washed dishes, and he never forgot it.

Jensen started out washing dishes at Denny's. Now his chips power AI all over the world. Do every job well, even the small ones.

Alex: The Thinker Who Learned Differently



When Alex Karp was a boy, reading was hard for him.

Alex has dyslexia. Dyslexia makes letters and words hard to read.

Something was hard. And Alex found a way.

Alex was born in 1967 in New York City. He grew up in Philadelphia with his younger brother. His dad was a children's doctor. His mom was an artist.

Alex says dyslexia shaped the way he thinks. Here is what he said

"There is no playbook a dyslexic can master. And therefore we learn to think freely."

A playbook is a set of rules that tells you what to do. Alex could not just follow the rules everyone else used. So he learned to think in his own way.

Alex grew up and kept learning. He studied big ideas. This is called philosophy. Philosophy asks big questions, like "What is fair?" and "What is true?"

Then he went to law school at Stanford, to learn about laws. After that he went all the way to Germany, a country in Europe. There, in 2002, he earned a PhD. A PhD is one of the highest degrees you can earn in school.

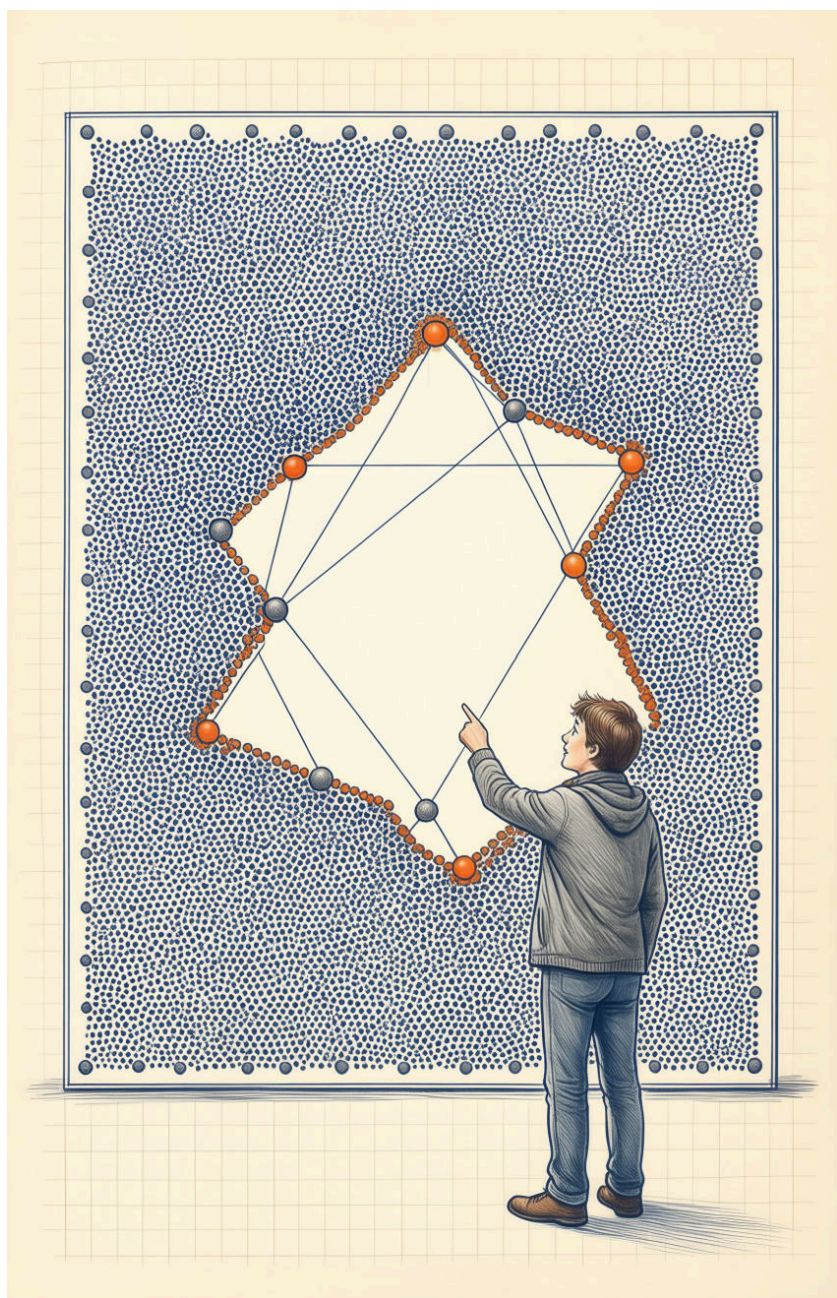
The Big Idea

In 2003, Alex and some friends started a company called Palantir. Alex has led it ever since. That is 23 years!

Palantir makes software. Software is the programs that tell a computer what to do.

Imagine a giant board covered in dots. Thousands and thousands of dots. Each dot is one piece of information. Alone, one dot does not tell you much.

Palantir's software connects the dots. Then people can see what matters.



What Palantir Builds Now

Palantir has three big programs.

Gotham helps governments. It helps armies understand huge piles of information.

Foundry helps businesses and hospitals. It helps them see what is going on and what to fix.

AIP uses artificial intelligence, or AI.

Builders Right Out of High School

Most big companies want workers who went to college. Palantir tried something different.

In 2025, Palantir started a program called the Meritocracy Fellowship. Meritocracy means you get picked for what you can do, not for where you went to school.

The program is for kids who just finished high school. Instead of going to college, they go to work at Palantir. And they get paid!

First, they learn. They study history and big ideas, a little like Alex did.

Then they build. They work on real projects with Palantir's software, right next to grown-up engineers. An engineer is a person who designs and builds things.



The ones who do great work can get a job at Palantir. No college needed.

How Alex Works

Alex trains like an athlete. He goes cross-country skiing for hours. Cross-country skiing is skiing across flat, snowy land, not just down hills.



Every day, he also does qigong. Qigong is a set of slow, gentle moves with deep breathing.

Wow!

In three months in 2026, Palantir sold almost \$2 billion of its software.

That is nearly double what it sold the year before.

Alex has led Palantir for 23 years.

Builder's Toolbox: thinking freely. When the usual way did not work for Alex, he found his own.

Reading was hard for Alex. So he learned to think in a new way. If something is hard for you, look for your own way through it.

You Can Be a Builder Too



Palmer, Elon, Jensen and Alex were all kids once. Just like you.

Palmer had half a garage. Elon had a computer and an idea for a space game. Jensen had a stack of dirty dishes. Alex had words that were hard to read.

None of them started big. Each one started with what he had.

So what do builders have in common?

They ask, "How does this work?" They keep learning, even when school is done. They work hard, even when the job is small. And when something is hard, they find their own way.

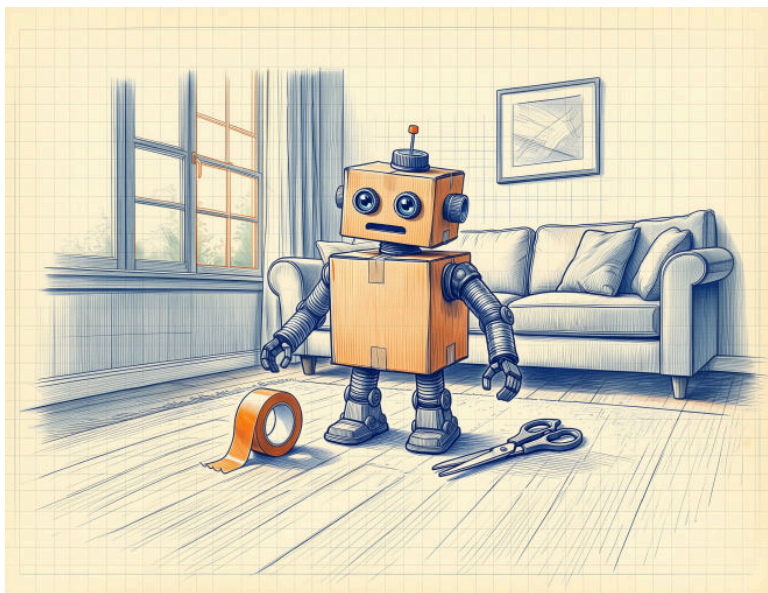
You can do every one of those things.

Here is the best part. You can start today. Remember the secret from the start of this book? You do not have to wait until you are grown up. You can just do things.

Try one of these this week, with a grown-up.

Take apart an old toy. Look at every piece. Why is it made that way?

Build a robot out of cardboard boxes. Give it arms. Give it a name.

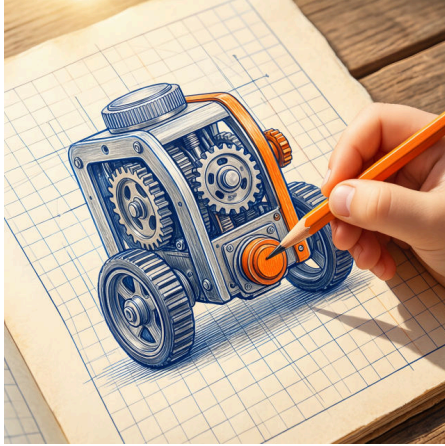


Make a tiny computer game. Elon made his first game when he was 12.

Find something broken and fix it. Palmer fixed broken phones to pay for his parts.

Then look around you. What is a problem you see? What could you build to make it better?

Draw your idea. Write it down. Then start building.



Builder's Toolbox: everything in this book. It is yours now.

The world is full of things nobody has built yet. One of them is waiting for you.



A true picture book for 2nd graders about four builders: Palmer Luckey (Anduril), Elon Musk (SpaceX, Tesla), Jensen Huang (Nvidia) and Alex Karp (Palantir). Each gets a chapter: where they grew up, the first thing they built, how hard they worked, and the amazing things their companies make now (fall 2026). Ends with a chapter telling the reader they can be a builder too.